

CS105 (DIC on Discrete Structures)

Problem set 9

- Apart from things proved in lecture, you cannot assume anything as “obvious”. Either quote previously proved results or provide clear justification for each statement.
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Part 1

1. Show that if there are 30 students in a class, at least two have last names beginning with the same letter.
2. In CS105 this semester there are 205 registered students. They have to be given grades, which must be of course one of following: AA, AB, BB, BC, CC, CD, DD, FR.
 - (a) Can we be sure that at least 25 students have got the same grade? Why or why not?
 - (b) What is the minimum number of students required in a course to be sure that at least 30 must have the same grade?
3. Three children, Aamir, Akbar, and Anthony, decide to share 11 sweets for Diwali. They wonder how many ways they could split the sweets up provided that none of them receive more than 4 sweets (someone receiving no sweets is for some reason acceptable to these kids!).
4. In any group of 7 people, show that there must be a group of 3 people who are mutual friends (that is, each pair of them are friends), or a group of 3 people who are mutually unknown to each other. You may assume that between any two people in the group, they are either friends or don't know each other.
5. Let A be a subset of the set $\{1, 2, 3, \dots, 200\}$ such that $|A| = 101$. Then, using Pigeon-hole Principle, prove that A must contain x, y such that $x \neq y$ and x divides y . Generalize this to show that among any $n + 1$ positive integers not exceeding $2n$ there must be an integer that divides one of the other integers ($n \in \mathbb{N}^{>0}$).

Part 2

6. Given five points on the interior of a square with side length 2cm, prove using Pigeon-Hole-Principle that there must exist two of the points that are < 1.5 cm apart. What is the smallest real number you can replace 1.5 with so that this statement is still true?
7. Let x be an irrational number. Show that for some positive integer j not exceeding the positive integer n , the absolute value of the difference between jx and the nearest integer to jx is less than $\frac{1}{n}$. What about if j can go up to $n - 1$? Find a counter-example of n, x ($n > 1$) or prove that a j always exists in this case as well.
8. A derangement is a permutation of objects that leaves no object in its original position. Let D_n denote the number of derangements of $\{1, \dots, n\}$. Thus, D_n is the number of ways in which letters numbered 1 to n can be put into boxes numbered from 1 to n such that for any i , the letter i is not in box i .
 - (a) Write a recurrence for D_n the number of derangements.
 - (b) Obtain a closed form solution for D_n in terms of n , using Principle of Inclusion Exclusion (PIE).
 - (c) What is $\lim_{n \rightarrow \infty} \frac{D_n}{n!}$?
9. A bijection $f : \{1, \dots, 2n\} \rightarrow \{1, \dots, 2n\}$ is said to be *cool* if there exists $i \in \{1, \dots, 2n - 1\}$ such that $|f(i) - f(i + 1)| = n$. Prove that, for each n ($n \in \mathbb{N}^{>0}$), more than half the bijections from $\{1, \dots, 2n\}$ to $\{1, \dots, 2n\}$ are *cool*.